A Candidate Lean 4 Statement of the Yang–Mills Existence and Mass Gap Problem

Adversarial AI Review and a Research Programme
(No Proof Is Claimed)

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Statement files unchanged since 6 November; pre-registration claim withdrawn after the release of OpenAI's Lean code on 10 November.

Abstract

On 26 October 2026 OpenAI announced an AI-generated proof of the existence of four-dimensional quantum Yang–Mills theory, formalized in Lean, and placed it with a shepherding committee whose first stated objective is to check that "the definitions and problem statement correspond with those which are standard in physics and mathematics". On 10 November 2026 the committee published OpenAI's Lean code, with an assessment that its announcement describes as a validation that the code proves the existence of pure Yang–Mills theory [1]. That release came after the statement, its review and the research library reported here were completed on 6 November; this version examines neither the code nor the assessment. This paper does *not* prove existence or the mass gap. It contributes four things that are independent of OpenAI's material. (i) A candidate formal statement, in Lean 4 with Mathlib, of the Clay Millennium problem "Yang–Mills existence and mass gap" and of an existence-only variant. The statement uses the Osterwalder–Schrader axioms with OS II's linear growth condition, a Euclidean formulation of the mass gap, and identifies the theory as Yang–Mills through a continuum-limit clause for Wilson's lattice gauge theory, a full-field-content clause for the 21 components of $\operatorname{tr} F_{\mu\nu}F_{\kappa\lambda}$, and a short-distance asymptotic-freedom fingerprint. (ii) Machine-checked sanity and non-vacuity results: the trivial theory satisfies the axioms but not the non-triviality clause; the lattice partition function is positive; and $SU(N)$, $N \geq 2$, satisfies the hypotheses of the general statement ($SU(N)$ is connected and its Lie algebra, computed in Lean to be $\mathfrak{su}(N)$, is simple). All of these depend only on Lean's three standard axioms. (iii) An adversarial-review methodology, executed with AI agents, that found a critical defect in the first draft (the cluster axiom admitted the zero vector, which collapses the Hilbert space and would have made the open theorems provably false) and four major fidelity defects, two fixed and two only partly (short-distance agreement is imposed only through a scalar two-point fingerprint, and the dependence on Wilson's lattice regularization remains); both residues are recorded as deviations from the Clay text. (iv) A research library compiled and cross-checked by AI agents (11 notes; 666 bibliography entries whose existence and metadata were checked, a few flagged as unconfirmed) and a research programme for an independent proof, informed by the published record of OpenAI's agent-swarm attack on Navier–Stokes and its public Lean repositories, with explicit subjective odds, stop rules and an independence policy. The statement's main design elements also appear in an earlier candidate statement by Rémondière [2]. AI systems wrote all of the mathematics, Lean code and prose under the author's direction; no human expert has reviewed the work; Section 9 gives a full disclosure.

Contents

1 Introduction **2**

2 The problem **4**
2.1 What Jaffe and Witten ask for . . . 4
2.2 Osterwalder–Schrader axioms and linear growth . . . 4
2.3 Status . . . 5

3 A candidate formal statement in Lean 4 **5**
3.1 Design choices . . . 5
3.2 Axioms . . . 6
3.3 The mass gap . . . 6
3.4 Identifying the theory as Yang–Mills . . . 6
3.5 The targets . . . 7
3.6 Milestone ladder . . . 8
3.7 Deviations from the Clay text . . . 8

4 Machine-checked results **8**

5 Adversarial review of the statement **9**
5.1 Method . . . 9
5.2 Findings . . . 9

6 Research landscape **10**

7 A research programme **12**
7.1 What the premise implies . . . 12
7.2 Targets and tracks . . . 13
7.3 AI operating model . . . 13
7.4 Honest odds and stop rules . . . 14
7.5 Independence and ethics . . . 14

8 Limitations **15**

9 Disclosure of AI use **15**

10 Data and code availability **17**

1 Introduction

The Clay Mathematics Institute's Millennium problem on Yang–Mills theory, described by Jaffe and Witten [3], asks for a proof that for every compact simple gauge group G a non-trivial quantum Yang–Mills theory exists on $\mathbb{R}^4$ and has a mass gap $\Delta > 0$, where existence includes "axiomatic properties at least as strong as" those of Streater–Wightman [4] and Osterwalder–Schrader [5, 6]. Douglas's 2004 status report [7] recorded the state of the art as Bałaban's renormalization-group programme for lattice gauge theory [8–12] and the infrared-cutoff construction of Magnen, Rivasseau and Sénéor [13]; a 2026 review is [14].

The 2026 situation. On 8 September 2026 OpenAI announced a Lean-formalized proof of finite-time blowup for the three-dimensional Navier–Stokes equations with smooth forcing, produced by a large multi-agent system [15–18]; the scope of that result relative to Fefferman's problem statement [19] is disputed because it uses the forced alternative [20]; Constantin, Ignatova and Vicol show that constructions with its key features cannot have a real-analytic force [21]. On 26 October 2026 OpenAI announced, among other results, an existence proof for pure Yang–Mills theory in four dimensions, as part of a claimed construction of certain interacting gauge theories with fermions; the mass gap is not claimed and no gauge group is named

publicly [22–25]. The natural-language proofs ("the full 1147-page LLM proofs" according to the committee, covering an unstated subset of the claimed results) and the Lean formalization of pure Yang–Mills existence are held by a shepherding committee (Douglas, Gross (chair), Jaffe, Klebanov, Nekrasov), which states that Bałaban's ideas were "among the important elements" and that confinement and the mass gap remain open [26–28]. As of 6 November 2026, when the statement, its review and the research library reported here were completed, none of this material was public. On 10 November 2026 the committee published the Lean code that OpenAI submitted to it on 28 October, together with an assessment [1]. The committee's announcement calls the assessment a "Validation of the OpenAI Yang–Mills construction (Lean proof)" and describes it as summarizing the committee's checks that the code proves the existence of pure Yang–Mills theory according to the problem statements standard in physics and mathematics; that is the committee's conclusion, not one examined here. The announcement covers only the Lean code and the assessment. GitHub lists the repository holding OpenAI's code as created on 5 November 2026; the committee announced that it would make it public on 10 November. Neither the code nor the assessment is examined in this version. Independently, Sussman and Alpöge have posted AI-generated, self-described tentative manuscripts continuing Bałaban's programme in three dimensions, with an unfinished four-dimensional part [29], and Aronow and Lopatto have posted AI-generated manuscripts claiming the continuum limit of three-dimensional $SU(2)$ Yang–Mills theory on a torus [30]. None of these claims has been refereed.

Why a formal statement matters now. A Lean proof certifies only the theorem that was stated, not the paper's argument: the Navier–Stokes Lean development reaches its headline theorem through intermediate lemmas that differ from the paper's [31]. Errors of meaning therefore concentrate in the statement. The maintainers of the lean-dojo collection mark its Yang–Mills statement as incomplete and "not a valid target", noting that a two-dimensional toy model satisfies it [32]. An earlier AI-written candidate statement by Rémondière, deposited on Zenodo on 25 October 2026 [2], already combines a Wilson-lattice-limit variant with an axiomatic variant carrying a Yang–Mills "fingerprint", uses the 21 components of $\mathrm{tr}\, F_{\mu\nu}F_{\rho\sigma}$ as field content, and concedes that its gap covers only the C-even sector they generate; like the present work, it was written by Claude agents under its author's direction. Our statement shares these main design elements. The library note that reviewed it (note 01) was written before our version 0.1 was reviewed, and our version 0.2 additions were made by reviewer agents who had read that note. As of 6 November 2026 the formal-conjectures collection [33] contains no merged Yang–Mills statement (an open pull request proposes one). The shepherding committee lists statement fidelity as its first objective [26], and its 9 November announcement reports a positive result of checks of that kind [1]. An AI-reviewed candidate statement written independently of OpenAI's code, with published file hashes, is useful as a point of comparison whatever the status of that proof. It is *not* a pre-registration. The statement files of version 0.2.0 were completed on 6 November 2026, four days before the committee published OpenAI's Lean code, but that date rests on the author's account and on local Git commits that carry no independent timestamp. Those commits were squashed into a single commit before first publication, to remove local file paths, so they are not public: the published history begins on 10 November 2026, and this deposit was made after the release. The author states that the statement files were not changed after the release and that OpenAI's Lean code was not consulted in writing them; neither claim can be checked independently. What the deposit does fix is the content: the SHA-256 hashes of the statement files are listed in Section 10, and any later version will be published with a diff against 0.2.0 and the reason for each change.

Premise and scope. This work was undertaken under a working premise set by the author: *assume that OpenAI's existence proof and its Lean formalization are correct, and aim to produce*

an independent proof and formalization. The premise is not evaluated here. Section 7 explains what it does and does not imply for our formal targets.

Contributions.

1. A candidate Lean 4/Mathlib statement (Section 3) of `ClayYangMills`, its existence half `ClayExistence`, an $SU(N)$ existence statement along the asymptotic-freedom trajectory (`SUNExists`), and an ordered ladder of weaker milestone statements (implied by `SUNExists`, with the implications proved in Lean). Its main design elements also appear in the earlier candidate of [2].
2. Fully machine-checked sanity and non-vacuity results (Section 4).
3. An adversarial statement-review methodology and its findings (Section 5).
4. An AI-compiled, AI-cross-checked research library and an explicit research programme with odds, stop rules and an AI operating model informed by the published record of 2026's large-scale AI attacks on Millennium problems (Sections 6 and 7).

This paper contains no proof of existence or of a mass gap, and does not claim one.

2 The problem

2.1 What Jaffe and Witten ask for

The official description [3] asks for a quantum field theory "in the above sense" (Wightman/OS axioms [4–6]; see also [34, 35]) satisfying, in addition:

1. **Field content.** Local quantum fields correspond to gauge-invariant local polynomials in the curvature F and its covariant derivatives, e.g. $\operatorname{tr} F_{ij}F_{kl}(x)$; the correspondence is not canonical (a footnote notes renormalization subtleties).
2. **Short distances.** Correlation functions agree at short distances with asymptotic freedom [36, 37] and textbook renormalized perturbation theory, including a stress tensor and an operator product expansion.
3. **Mass gap.** $H\Omega = 0$, the spectrum of H lies in $[0, \infty)$, and H has no spectrum in $(0, \Delta)$; the supremum m of such Δ must be finite.
4. **No weak existence.** Existence by compactness alone is not a solution unless the axioms and the gap are established for the limit.
5. **A Hilbert space.** "Establishing a quantum mechanical Hilbert space is part of the solution."

2.2 Osterwalder–Schrader axioms and linear growth

Osterwalder and Schrader [5] gave Euclidean axioms (E0)–(E4) for Schwinger distributions S_n on $\mathcal{S}_0(\mathbb{R}^{4n})$ (Schwartz functions vanishing with all derivatives on all diagonals): temperedness, Euclidean covariance, reflection positivity, symmetry, and cluster property. A lemma in [5] is incorrect, and OS II [6] replaced temperedness by the *linear growth condition*

$$|S_n(f)| \le \sigma_n \, |f|_{ns}, \qquad \sigma_n \le \alpha \, (n!)^{\beta}, \tag{E0$'$}$$

for a fixed Schwartz norm index s, under which reconstruction of a Wightman theory holds. The Glimm–Jaffe measure axioms [34] are not suitable here. A short argument of ours (library note 01, §2.6; not a published result) shows that their regularity axiom OS1 is incompatible with any field of scaling dimension greater than two, and so with the expected $|x|^{-8} \log^{-2}(1/|x|)$ behaviour of $\operatorname{tr} F^2$.

2.3 Status

Table 1 summarises the status on 6 November 2026; details and sources are in the library notes (Section 10).

Item	Status
4D existence (gauge groups not stated)	**Claimed** by OpenAI for pure Yang–Mills theory; gauge groups, axioms and Lean statement not public on 6 November; Lean code released by the committee on 10 November, after this work, with an assessment that the committee's announcement describes as validating it; neither is examined here [1, 22, 26, 27].
4D mass gap	**Open**: not claimed by OpenAI [22, 23]; the committee states that it remains open [26].
3D Yang–Mills continuum	No refereed construction; AI-generated claims [29, 30].
2D Yang–Mills	Constructed [38–42]; Langevin dynamics and universality [43, 44].
4D lattice mass gap, continuous non-Abelian G	Proved only at strong coupling [45–47]; never in the weak-coupling continuum regime. (Finite G: correlation decay at weak coupling [48]; 3D $U(1)$, Villain action: area law at all couplings [49].)

Table 1: Status of the problem and its relatives (6 November 2026, with the 10 November release noted).

3 A candidate formal statement in Lean 4

The statement is a Lean 4 [50] project built on Mathlib [51] (toolchain and Mathlib tag `v4.36.0-rc2`, Mathlib commit `9d581df`). It imports only Mathlib, contains definitions and proofs only, and confines every `sorry` to one file of open problems. About 1,700 lines are spread over `OS.lean` (axioms), `Lattice.lean` (lattice gauge theory), `SU.lean` ($SU(N)$), `Statement.lean` (targets), `OpenProblems.lean`, `Sanity.lean`, `NonVacuity/*.lean` and `Audit.lean`.

3.1 Design choices

1. **Euclidean OS form.** OS form is closest to every constructive route; OS II yields a Wightman theory.

2. **Multilinear Schwinger functions.** For fields φ_i, $i \in \iota$, transforming under a real representation ρ of $O(4)$, the n-point function is a continuous multilinear form $S_n(f_1, \dots, f_n) = \langle \varphi(f_1) \cdots \varphi(f_n) \rangle$ on $f_k \in \mathcal{S}(\mathbb{R}^4, \mathbb{C}^\iota)$. This avoids building tensor products of Schwartz functions.

3. **Disjoint supports.** Composite-operator correlations have scheme-dependent contact terms. Following OS's use of $\mathcal{S}_0$, every axiom except growth is imposed only on tuples with pairwise disjoint supports; reflection positivity is imposed on products supported in ordered positive-time slabs, the multilinear analogue of OS's $\mathcal{S}_+$.

4. **Constructed operations, not relations.** Euclidean motions, translations and the reflection-conjugation Θ are built as continuous linear maps from Mathlib's `SchwartzMap.compCLMOfContinuousLinearEquiv`, `compSubConstCLM` and `postcompCLM`. A purely relational definition would risk vacuity if no related function existed.

3.2 Axioms

Write $\Theta f = \overline{\rho(\theta)^{\mathsf{T}} f(\theta\,\cdot)}$ with θ time reflection, and for finite linear combinations $F = \sum_j c_j\,\varphi(f_{j,1})\cdots\varphi(f_{j,n_j})$ and $G = \sum_l d_l\,\varphi(g_{l,1})\cdots\varphi(g_{l,m_l})$ of ordered-positive monomials define the reflection-positivity form

$$B(F,G) = \sum_{j,l} \overline{c_j}\, d_l\; S_{n_j+m_l}(\Theta f_{j,n_j}, \dots, \Theta f_{j,1}, g_{l,1}, \dots, g_{l,m_l}).$$

`OSAxioms` consists of:

- normalization ($S_0 = 1$);
- an equivalent form of (E0′): for each n, S_n agrees on disjointly supported products with a distribution T_n on $\mathcal{S}((\mathbb{R}^4)^n)$ obeying $|T_n(F)| \le \alpha(n!)^\beta |F|_{ns}$ (equivalent to OS II's condition on $\mathcal{S}_0$ by Hahn–Banach and density of disjointly supported products, with norm-convention constants absorbed into $(n!)^\beta$; this equivalence is argued, not formalized);
- reality;
- Euclidean covariance under $f \mapsto \rho(R^{-1})^{\mathsf{T}} f(R^{-1}(\cdot - a))$ for all $R \in O(4)$, $a \in \mathbb{R}^4$;
- permutation symmetry;
- reflection positivity $B(F,F) \ge 0$;
- the cluster property $S(\Theta A^* \cdot T_{\lambda a} B) \to S(\Theta A^*)S(B)$ as $\lambda \to \infty$, for every *non-zero* spatial a.

3.3 The mass gap

With the truncated form $B_T(F,G) = B(F,G) - B(F,1)B(1,G)$ and Euclidean time translation T_t:

Definition 3.1 (`HasMassGap`)**.** $\Delta > 0$ and, for all ordered-positive F, G and all $t \ge 0$, $|B_T(F, T_t G)| \le e^{-\Delta t}\, B_T(F,F)^{1/2} B_T(G,G)^{1/2}$.

Proposition 3.2. *After OS reconstruction,* `HasMassGap` Δ *is equivalent to* $\|e^{-tH}(1-P_\Omega)\| \le e^{-\Delta t}$ *for* $t \ge 0$*, i.e. the spectrum of the Hamiltonian on* $\Omega^\perp$ *lies in* $[\Delta, \infty)$ *and the vacuum is unique.*

Sketch. Reconstruction maps ordered-positive F to vectors $\hat{F}$ in the OS Hilbert space $\mathcal{H}$ generated by the fields, with $\langle \hat{F}, \hat{G} \rangle = B(F,G)$ and $\widehat{T_t G} = e^{-tH}\hat{G}$, $H = H^* \ge 0$. These vectors are dense because finite sums of products of slab-supported test functions are dense in OS's $\mathcal{S}_+$; this density step, like the proposition itself, is argued but not yet formalized. Then B_T is the inner product of the projections onto $\Omega^\perp$, and the inequality on a dense set is the stated operator bound. The spectral theorem converts the operator bound into the spectral statement. □

With Lorentz covariance (from OS II), this is the Jaffe–Witten gap. The finiteness of m follows from non-triviality.

3.4 Identifying the theory as Yang–Mills

The OS axioms, non-triviality and a gap are satisfied by theories that are not Yang–Mills (for example Wick powers of a massive free field). Four clauses pin the theory down.

Non-triviality (`NonTrivial`). The fields are centred; some two-point function and some fourth cumulant are non-zero at separated points; and some state orthogonal to the vacuum has positive norm.

The lattice clause (`IsWilsonLimit`). For a compact group G with a faithful continuous unitary representation $\pi : G \to U(d)$, consider the periodic lattice $(\mathbb{Z}/(L+1))^4$, link variables U, plaquettes $U_{\mu\nu}(x)$, and Wilson's action [52]

$$S_\beta(U) = \beta \sum_x \sum_{\mu<\nu} \Big(1 - \tfrac{1}{d} \operatorname{Re}\,\operatorname{tr}\, \pi(U_{\mu\nu}(x))\Big),$$

with the Gibbs measure built from normalized Haar measure. The lattice field strength $F_{\mu\nu}(x)$ is the traceless Hermitian part of $(\pi(U_{\mu\nu}) - \pi(U_{\mu\nu})^\dagger)/2i$, and $T_{\mu\nu\kappa\lambda}(x) = \operatorname{Re}\,\operatorname{tr} F_{\mu\nu}(x) F_{\kappa\lambda}(x)$. Smeared fields are Riemann sums $a^4 \sum_x f(ax)\, Z\,(T(x) - \langle T(x)\rangle)$ with renormalization matrices Z valued in the 21-dimensional space of tensors with the symmetries of $\operatorname{tr} F_{\mu\nu}F_{\kappa\lambda}$. S is a *Wilson limit* if, along some sequence with $a_k \to 0$, $a_k(L_k+1) \to \infty$ and $\beta_k \to \infty$, the lattice Schwinger functions converge to S on compactly supported test functions with disjoint supports. O(4) acts on the fields by $R \otimes R \otimes R \otimes R$; the Lean representation is a monoid homomorphism built from Kronecker products.

Full field content (`FullFieldContent`). Every non-zero direction v in the 21-dimensional space yields a non-zero state $\varphi_v(h)\Omega$. Without this clause, the existential renormalization matrices could project the theory onto a single channel such as $\operatorname{tr} F^2$; this is a finding of the review in Section 5.

Short-distance asymptotic freedom (`ShortDistanceAF`). In the scalar channel ($v_{\mu\nu\kappa\lambda} = \delta_{\mu\kappa}\delta_{\nu\lambda} - \delta_{\mu\lambda}\delta_{\nu\kappa}$), for compactly supported h_1, h_2 with disjoint supports and dilations $D_\lambda h(x) = \lambda^{-4} h(x/\lambda)$, we require for some $C > 0$

$$\lambda^8 \log^2(1/\lambda)\; S_2(D_\lambda h_1, D_\lambda h_2) \;\longrightarrow\; C \iint h_1(x)\, h_2(y)\, |x-y|^{-8}\, dx\, dy \qquad (\lambda \to 0^+).$$

The motivation is perturbative. The renormalization-group-invariant trace-anomaly operator $\Theta = (\beta(g)/2g) F^a_{\mu\nu} F^a_{\mu\nu}$, with $\beta(g) = \mu\, dg/d\mu = -b_0 g^3 - b_1 g^5 + \cdots$ [53], has at leading order correlator $\propto \bar g^4(1/|x|)\, |x|^{-8}$, where $\bar g(\mu)$ is the running coupling at scale μ; with $\bar g^2(\mu) \sim 1/(2b_0 \log(\mu/\Lambda))$ this gives a $\log^{-2}$ factor absent in free field theory. The clause therefore excludes free massless gluons even without a gap. It is a *requirement*; its validity for Yang–Mills is expected from perturbation theory but is not proved.

Asymptotic scaling (`AsymptoticScaling`, $SU(N)$ only). With $\beta = 2N/g_0^2$, $b_0 = 11N/(3(4\pi)^2)$ [36, 37] and $b_1 = 34N^2/(3(4\pi)^4)$ [54, 55], $\beta_k/(2N) - 2b_0 \log(1/a_k) - (b_1/b_0)\log\log(1/a_k)$ converges as $k \to \infty$.

3.5 The targets

```
def ClayYangMills : Prop :=
  ∀ (d₀ : ℕ) (G : Type) [Group G] [TopologicalSpace G] [IsTopologicalGroup G]
    [CompactSpace G] [ConnectedSpace G] [MeasurableSpace G] [BorelSpace G]
    (P₀ : GaugeGroup d₀ G),
    IsSimpleMatrixLieAlgebra P₀.lieAlgebra → ExistsWithMassGapFor G

def IsYangMillsTheoryAlong P S a β : Prop :=
  OSAxioms tensorRep S ∧ NonTrivial tensorRep S ∧ FullFieldContent S ∧
    ShortDistanceAF S ∧ IsWilsonLimit P S a β
```

`ExistsWithMassGapFor G` asserts that, for some faithful unitary representation of G, there is a Yang–Mills theory in this sense with `HasMassGap` Δ for some Δ. `ClayExistence` drops the

gap. `SUNExists N` uses the fundamental representation of $SU(N)$ and adds `AsymptoticScaling`. `SUNExistsWithMassGap N` uses the fundamental Wilson action and requires the gap. The docstrings of `SUNExists` and `su_yang_mills_exists` call it the shape or scope of OpenAI's announced result. That wording predates the 10 November release, was not checked against OpenAI's Lean code, and is left unchanged because the statement-file hashes are fixed (Section 10).

3.6 Milestone ladder

`SUNExists` bundles two genuinely infrared clauses, full $O(4)$ covariance and the cluster property. Given the other axioms, the cluster property is equivalent to uniqueness of the vacuum; since `IsWilsonLimit` uses periodic tori with no boundary conditions, the periodic limit itself must have a unique vacuum. We therefore define weaker rungs: `SUNExistsM7a` (hypercubic covariance, no cluster clause) and `SUNExistsM7b` (full $O(4)$, no cluster clause), and prove `SUNExists` $\to$ M7b $\to$ M7a in Lean. Only `SUNExists` (and `ClayExistence` for general G) corresponds to the existence part of the Clay problem.

3.7 Deviations from the Clay text

The Clay text is informal in places, so no formal statement can be proved equivalent to it. Table 2 records each deviation and its direction.

Aspect	Direction	Consequence
Gap only on the cyclic space of $\mathrm{tr}\, FF$ (a subspace of the C-even sector)	weaker	a full proof must add C-odd and higher-dimension fields
Wilson-lattice origin required	stronger/different	other regulators need a universality theorem; Douglas [7] notes the lattice is "not essential"
AF fingerprint in the scalar two-point function only	weaker	no stress-tensor or OPE clause
Subsequential limits accepted	choice	axioms and gap are required of the limit, so JW's "no weak existence" is met
E0′ rather than Glimm–Jaffe OS1	OS II's own condition	OS1 is incompatible with the expected singularity of $\mathrm{tr}\, F^2$ (our argument; see Section 2)
Representation chosen existentially	equivalent intent	a fundamental-representation variant is kept separately

Table 2: Deviations of the candidate statement from the Clay problem description.

4 Machine-checked results

Table 3 lists the proved theorems. For each, `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The open problems additionally depend on `sorryAx`.
Open (`sorry`): `clay_yang_mills`, `clay_existence`, `su_yang_mills_exists`, `su_yang_mills_mass_gap`. One classical but unformalized obligation, `os_nonTrivial_fullField_consistent`, asserts that the OS axioms are consistent with `NonTrivial` and `FullFieldContent` (witness: Wick-ordered composites of the free Maxwell field with values in $\mathfrak{su}(N)$). Consistency of the full Yang–Mills conjunction, including `ShortDistanceAF` and the lattice clause, is the open problem itself.

Theorem	Content
`trivial_osAxioms`, `trivial_massGap`	The trivial theory ($S_0 = 1$, $S_n = 0$) satisfies every OS axiom and has a gap: the axioms as formalized are consistent.
`trivial_not_nonTrivial`	… but it is not `NonTrivial`: the clause does real work.
`exists_spatial_ne_zero`	Regression guard: the cluster axiom quantifies over a non-empty set.
`partitionFn_pos`, `expect_const`	For second-countable G (which includes every compact group with a faithful continuous unitary representation), the Wilson partition function is positive, so lattice expectations are genuine averages; $\langle c \rangle = c$.
`isCompact_specialUnitaryGroup`	$SU(N)$ is compact (not previously in Mathlib).
`su_connected`	$SU(N)$ is connected (via `su_pathConnectedSpace`): $U(N)$ is shown path-connected from Mathlib's `Unitary.joined`, and a determinant-correcting retraction maps it onto $SU(N)$.
`suGauge_lieAlgebra`	The Lie algebra $\{X : e^{tX} \in SU(N)\ \forall t\}$ equals $\mathfrak{su}(N)$; uses $\det e^Y = e^{\mathrm{tr}\, Y}$ for skew-Hermitian Y, proved via the spectral theorem.
`su_isSimple`	$\mathfrak{su}(N)$, $N \geq 2$, is a simple real Lie algebra.
`su_mass_gap_of_clay`	`ClayYangMills` implies the conclusion for $SU(N)$: the general statement is not vacuous.
`SUNExists.toM7b`, `SUNExistsM7b.toM7a`	The milestone ladder is ordered.

Table 3: Fully proved results in the statement package.

5 Adversarial review of the statement

5.1 Method

Version 0.1 of the statement was reviewed by six AI reviewer agents (separately prompted, context-isolated instances of the same model), each with a distinct lens:

- fidelity to OS I/II;
- mass gap and non-triviality;
- whether the lattice clause pins down Yang–Mills;
- Lean technicalities and junk values;
- comparison with the Clay text and with public formalizations [56–59];
- a red team instructed to prove or disprove the open theorems cheaply.

Reviewers worked read-only on the statement but could compile scratch Lean files, so exploits were demonstrated, not argued. Each finding then went to a separate verifier agent (same model) instructed to refute it, defaulting to "refuted" when in doubt. Of 14 verified findings, 13 survived and 1 was refuted. Compiling exploits in Lean, together with the non-vacuity witnesses of Section 4, is in the spirit of the mechanical vacuity checks of Mei and Douglas [60], who combine LLM-based semantic review with mechanical numerical and vacuity checks and report that LLM judges miss errors those checks catch; our verification of findings, by contrast, was LLM-judged.

5.2 Findings

- **Critical: the cluster axiom admitted $a = 0$.**
  - Found separately by three reviewer agents (same model, different lenses).

  - The condition "a spatial" was encoded as $a^0 = 0$, which the zero vector satisfies. At $a = 0$ the sequence is constant, so the axiom forced exact factorization $S(\Theta A^* B) = S(\Theta A^*)S(B)$. That collapses the OS Hilbert space to $\mathbb{C}\Omega$.
  - Consequences: `HasMassGap` followed from the axioms, and the axioms together with `NonTrivial` were unsatisfiable. The open theorems were therefore *provably false*, and reviewers' Lean files proved the collapse.
  - Fixed by requiring $a \neq 0$. The reviewers' exploit files (kept in `scratch/`, outside the build) no longer compile, and `exists_spatial_ne_zero` shows that the corrected axiom is not vacuous; no automated regression test yet guards the fixes.
  - The trivial-theory sanity check could not have caught this, because the trivial theory factorizes exactly. Non-degenerate witnesses are required.

- **Major: growth weaker than E0′.** The first version bounded S_n by products of per-factor Schwartz norms of order $n \cdot s$. That bound follows from E0′ but is not known to imply it: the product-to-E0′ argument in the appendix to OS II needs per-factor norms of fixed order. So OS II's reconstruction theorem could not be invoked. Replaced by E0′ on configuration space.

- **Major: single-channel collapse.** Unconstrained renormalization matrices let a witness keep only $\operatorname{tr} F^2$. Lean lemmas showed that the Wilson-limit, reflection-positivity and gap clauses all survive such a projection. Fixed by `FullFieldContent` and a range condition on Z.

- **Major: missing short-distance requirement.** The JW requirement of agreement with asymptotic freedom had no formal counterpart. Added `ShortDistanceAF`, a scalar two-point fingerprint. The stress-tensor and OPE parts of the requirement remain unformalized and are recorded in Table 2; this finding is therefore also only partly fixed.

- **Major: method and representation dependence.** Requiring a Wilson-lattice limit makes the statement method-dependent and, in that respect, stronger than the Clay text, and universal quantification over faithful representations built lattice universality into the target. The representation was made existential. The Wilson-lattice requirement remains and is recorded in Table 2; a method-independent variant is future work. This finding is therefore only partly fixed.

- **Minor: documentation.** The renormalization matrices were described as "finite". Any non-trivial limit is expected to need $\|Z_k\| \to \infty$, like a_k^{-4} times channel-dependent powers of $\log(1/a_k)$ (perturbatively a_k^{-4} for the trace-anomaly scalar and $a_k^{-4} g_0^{-2}$ for the stress tensor).

6 Research landscape

We summarise the eleven library notes. Each claim there carries a label: established theorem, claimed but unverified, heuristic, disputed or open. The notes also label some claimed solutions as containing suspected errors; these are AI agents' assessments, not reviewed by any human expert, and should not be read as refereed judgements.

Constructive QFT and renormalization.

- Superrenormalizable models were constructed in the 1970s [34, 61].

- The massive Gross–Neveu model in two dimensions is generally cited as the first non-planar, non-hierarchical asymptotically free renormalizable theory to be constructed [62, 63]; planar φ_4^4 with the wrong sign was constructed earlier [64, 65].

- Modern renormalization-group methods include [66, 67], and cluster expansions [68].
- Aizenman and Duminil-Copin proved that scaling limits of critical four-dimensional Ising and (single-component, nearest-neighbour) lattice φ^4_4 models are Gaussian [69]. With the one-loop analysis of Coleman and Gross [70], this supports, but does not prove, the expectation that a non-trivial four-dimensional theory must be asymptotically free.

Bałaban's programme. In 1984–1989 Bałaban proved ultraviolet stability bounds (an upper bound on the partition function, and bounds on the effective densities) for lattice Yang–Mills on 3D tori, and on 4D tori conditionally on the running couplings staying small, using a gauge-covariant block-spin renormalization group. The work spans about 400 pages of *Commun. Math. Phys.* [8–12, 71–77].

- The 4D theorem is conditional on the running coupling staying small, and an announced companion result on the coupling flow was, according to secondary sources [78], never published. Checking this against the primary text is our first stop rule (SR1).
- According to these secondary accounts, correlation functions, convergence, infinite volume, the OS axioms and the gap are not treated in Bałaban's work.
- Dimock's expositions [78–82] are the only sustained, refereed modern re-derivation of Bałaban's method; they treat φ^4_3 and QED_3, not non-Abelian Yang–Mills. The 2026 AI-generated manuscripts [29, 30] are unrefereed.

Rigorous lattice results.

- Reflection positivity of Wilson's action, and confinement at strong coupling, are due to Osterwalder and Seiler [45, 46].
- Large-N and strong-coupling analyses [83–85], and exact random-surface representations of Wilson-loop expectations [86].
- Finite gauge groups [87–89], including correlation decay at weak coupling [48].
- Stochastic-analysis gaps at strong coupling [47], and area-law regimes [90, 91].
- Conditional reductions of confinement to correlation decay [92, 93].
- None reaches the weak-coupling regime of the continuum limit in four dimensions.

Stochastic quantization.

- Langevin dynamics, gauge-covariant state spaces and universality are established in two dimensions [42–44].
- In three dimensions, local solution theory and uniqueness of renormalization are established [94, 95], and Cao and Chatterjee [96, 97] supply a gradient-flow state space.
- A construction of the non-Abelian Yang–Mills–Higgs measure on the two-dimensional torus, for all fixed couplings, was posted on 2 November 2026 [98]. It is an unrefereed preprint, and its authors disclose that several lemmas were proved by AI models.
- In four dimensions the Yang–Mills Langevin equation is scaling-critical, which places it outside the locally subcritical class to which regularity structures [99] and related methods apply.
- Gradient-flow observables [100, 101], which are UV-finite to all orders of perturbation theory, are a natural observable class.

Mass-gap mechanisms and sanity filters. Any purported proof of a 4D gap must:

- fail for 4D $U(1)$, which has a massless Coulomb phase [102, 103];
- not prove confinement for finite groups;
- respect the $\theta = \pi$ anomaly [104].

Other relevant results:

- Rigorous confinement (an area law at all couplings) holds for 3D $U(1)$ lattice gauge theory with the Villain action [49].
- Controlled (physics-level, non-rigorous) semiclassical analyses exhibit gapped, confining regimes on compactified geometries [105]; their connection to $\mathbb{R}^4$ rests on an adiabatic-continuity conjecture.
- Finite-size uniqueness criteria exist [106].
- Computer-assisted renormalization group [107] is a sober calibration of what certified numerics can currently reach.
- Lattice Monte Carlo studies [108, 109] give strong numerical evidence for the gap.

AI-era claims (all unrefereed).

- OpenAI's announced results and repositories [58, 59, 110]. These include a claimed exponential decay of spin correlations at every positive temperature for the 2D nearest-neighbour $O(n)$ model with $n \geq 3$, with a Lean development covering finite free-boundary subgraphs [111].
- Sussman–Alpöge's 3D manuscripts [29].
- Aronow–Lopatto's 3D continuum limit [30].
- Craig's gauge-invariant lattice measure for chiral fermions [112].
- Public calls for release of OpenAI's material [113].

7 A research programme

7.1 What the premise implies

If OpenAI's claim is correct and, as reported, its proof is built on Wilson's lattice and Bałaban's renormalization group and covers fermions, then:

- *its* existence theorem is true;
- a Bałaban-type renormalization group on Wilson's lattice can be carried to infinite volume without a mass gap;
- the pipeline handles fermions.

The premise does *not* imply that our `SUNExists` is provable without new infrared mathematics. The cluster clause requires vacuum uniqueness for periodic limits. Limits of periodic tori can be non-trivial mixtures (as for the low-temperature Ising model, where every periodic limit is $\frac{1}{2}(\mu^+ + \mu^-)$), and the ergodic components of such a mixture need not be periodic limits; ruling this out for Yang–Mills needs infrared input. Nor does the premise say anything about the gap.

7.2 Targets and tracks

The programme climbs a ladder of frozen Lean statements, each adversarially reviewed and hash-recorded before proof search:

- finite-lattice foundations;
- a Gaussian $U(1)_4$ calibration rung;
- a 4D bounded-field multiscale renormalization group carrying the running coupling and a Θ-source;
- a conditional milestone in which Bałaban's published theorems enter as explicitly named hypotheses, as in OpenAI's repositories;
- finite-volume continuum limits;
- $T^3_\ell \times \mathbb{R}$ at small ℓ, the first genuine Hilbert space and gap;
- the rungs M7a/M7b/M7c.

The work is organized in five tracks:

- **A:** statement, bridges and release audit;
- **B:** demand-driven foundations in Lean;
- **C:** the ultraviolet analysis following Bałaban, starting from primary sources;
- **D:** infinite volume without a gap;
- **E:** the mass gap, as a capped research portfolio.

Within Track E, a desk analysis identified an obstruction. A finite-size criterion that is uniform over all volumes and boundary conditions cannot hold in four dimensions beyond bare coupling of order one. Take the time-like $SU(2)$ links of a height-one slab and freeze all other links to the identity; the conditional measure is then exactly the 3D $O(4)$ model at the same β, which has long-range order for large β. This is our heuristic argument, not a theorem (with generic frozen links one gets an $O(4)$ model in an $SO(4)$ background, which can be frustrated). It directs any certificate approach to cube-restricted criteria.

7.3 AI operating model

The design draws on the published record of OpenAI's Navier–Stokes swarm. That swarm reportedly used groups of agents with in-group messaging, a seeding stage from an Euler result, consolidation of insights across groups, and a separate formalization stage [15, 17, 18]. OpenAI researchers credit base-model strength over multi-agent organisation [114], and critics have documented the risks of scope loopholes [20, 21] and of Lean/paper drift [31]. The design also draws on OpenAI's Lean repositories: Comparator-gated challenges, honest-axiom patterns, conditional hypotheses, and metadata files [58, 59, 115–117]. Its principles are:

1. **The compiled Lean skeleton is the coordination substrate.** Each rung theorem is proved in Lean from a small number of named module contracts.
2. **Agents read contracts, not proofs.** Context is bounded.
3. **Mechanical checks first, model judgement second, the kernel last.** LLM skeptics are a rejection filter calibrated by planted errors, never an acceptance gate.

4. **Swarms only where the decomposition is unknown.**

5. **Provenance by construction.**

Blueprint tooling [118, 119] and Comparator [116] provide the gates.

7.4 Honest odds and stop rules

The programme adopts the subjective estimates of its feasibility review. At a sustained budget of about 10^{10} output tokens per year:

- an independent, Comparator-checked `SUNExists` is $\leq 1\%$ within 12 months and 3–8% within 36 months;
- a Lean bridge from OpenAI's theorem to ours is 60–80% likely within 3 months *if* their statement implies ours up to bookkeeping;
- the 4D mass gap is $< 0.1\%$ within 12 months and $< 0.5\%$ within 36 months at this budget, and $< 1\%$ within 36 months even at a Navier–Stokes-scale budget of about 10^{11} tokens per year.

Nine stop rules tie continuation to measurable checkpoints:

- choosing the clean-room/bridge/consume mix fourteen days after OpenAI's release;
- confirmation of the Bałaban gap in the primary text;
- calibrated derivation of b_0 in Bałaban's scheme;
- measured closure rate on a pilot formalization of Dimock's φ^4_3 analysis;
- canary recall of the review gates;
- spend caps;
- Track E caps;
- switching to verification if a complete public 4D manuscript appears;
- withdrawing the clean-room label from artefacts affected by a firewall breach.

7.5 Independence and ethics

Until OpenAI's release (10 November 2026), all work was clean-room by necessity; by policy it stays clean-room for fourteen days after the release (to 24 November), after which the author chooses among three modes:

- *clean-room*: no artefact derived from the release;
- *bridge*: our statement proved from theirs;
- *consume*: study the release and re-derive it.

Leakage controls and labelling rules accompany each mode. The programme commits to:

- contacting Sussman and Alpöge before duplicating their four-dimensional modules;
- respecting the committee's process;
- disclosing AI use;
- publishing any claimed proof, or any formal result beyond this candidate statement, only after Comparator checking and review by human experts, with no press releases. This paper itself is released without human expert review (Section 9), as a candidate statement and a plan, not as a result.

8 Limitations

- The statement is a *candidate*. It has been reviewed adversarially by AI agents, not audited by human experts in constructive QFT. Its fidelity to the Clay problem is argued, not proved; Table 2 lists the known deviations.
- The mass-gap clause covers only the cyclic space of the dimension-four fields.
- Consistency of OS + `NonTrivial` + `FullFieldContent` is asserted but not yet formalized.
- The library relies on secondary sources in important places. No paper of Bałaban's series was read in full (the publisher's site blocked automated access), so Section 6's account of his work rests on Dimock's expositions, other secondary accounts and an unvetted third-party transcription. OpenAI's blog posts [15, 22] returned HTTP 403 and were reconstructed from press and blog reports. Douglas's 2026 review [14] was not read. Bibliography entries that could not be confirmed are flagged.
- The Proposition of Section 3.3 and the equivalence of the multilinear axioms with OS's formulation are argued, not formalized.
- The research programme's odds are subjective. Its operating model is untested at the scale required.
- Nothing here independently evaluates OpenAI's claimed proof. Its Lean code, released on 10 November 2026 [1], was not examined for this version; a comparison of the two statements is left to a later version.

9 Disclosure of AI use

This disclosure follows, as far as they apply to an individual's project, the disclosure items that the Advisory Group on Mathematics and AI lists for AI-generated mathematics [120]: model, prompts, time, compute, and how AI was used. Those guidelines are addressed mainly to AI laboratories. Provided here: the model family and environment; the prompt given to every subagent (in the workflow scripts, `dev/workflows/`); and wall-clock times, tool calls and subagent token counts. Not included in this deposit: exact model versions (the Git commit messages in the GitHub repository carry an automatic co-author line naming only the model of the committing session), the author's natural-language instructions to the orchestrating session, that session's transcript and token count, summaries of the models' reasoning, and monetary cost.

Systems used. The work was carried out with Anthropic's Claude large language models, operated through the Claude Code agent environment. That environment includes a workflow facility that orchestrates many subagents in parallel, file, shell, web-search and web-fetch tools, and the Lean toolchain. Research subagents also used the alphaXiv paper-search and PDF question-answering service, a third-party tool whose answers may themselves be machine-generated. The models' training data predate the September 2026 events discussed here; all information about those events was obtained by live retrieval during the work.

What the human author did. The author:

- set the goal and the working premise;
- directed the work in natural language and answered the agents' questions;
- chose the repository visibility, publication route and licences;

- takes responsibility for publication.

The author did not hand-write the mathematics, the Lean code or the prose.

Human review. The author read the paper before deposit, including the passages revised on 10–11 November, but is not trained in constructive quantum field theory or in Lean, and could not independently check the mathematics, the physics or the Lean code. For these the paper relies on the Lean kernel (Section 4) and on the AI review described in Section 5. No human domain expert has reviewed this work. No text-overlap (plagiarism) check was run on the AI-generated text; quotations were checked against their sources by AI agents only. Corrections are welcome as issues on the GitHub repository.

What AI systems did. AI agents performed essentially all of the following:

- literature search and summarization;
- drafting of the library notes, the problem description, the plan and this paper (the orchestrating session wrote the Lean statement, its version 0.2 fixes, the problem description, the plan and this paper; subagents wrote the notes, reviews, critiques and non-vacuity proofs);
- design and writing of the Lean statement and all Lean proofs;
- adversarial review and verification of the statement, the notes, the plan and this paper.

Scale and structure (subagents only).

- **Research library.** 24 subagents (11 researchers, 11 adversarial citation and claim verifiers, an indexer and a completeness critic): about 9.1×10^6 tokens, 2,560 tool calls, about 2.0 hours.
- **Statement review.** 20 subagents (6 reviewers, 14 verifiers): about 3.6×10^6 tokens, 272 tool calls, about 24 minutes.
- **Plan critique.** 5 subagents: about 2.0×10^6 tokens, 194 tool calls, about 28 minutes.
- **Non-vacuity Lean proofs.** 3 subagents: about 0.6×10^6 tokens, 70 tool calls, about 12 minutes.
- **Paper verification.** 7 subagents (four citation checkers, a factual-claims auditor, a mathematics reviewer and a disclosure reviewer): about 2.0×10^6 tokens, 336 tool calls, about 22 minutes. Their findings, including the omitted citation of [2], were applied before deposit.
- **Pre-release audit.** 25 subagents (four auditors for privacy, reproducibility, metadata and claims, and 21 adversarial verifiers): about 4.2×10^6 tokens, 473 tool calls, about 19 minutes. It ran on 10 November 2026, after the release, with web access. It rebuilt the Lean project and the paper from a clean export of the then-current commit, and found that OpenAI's Lean code had been released, which made this paper's earlier pre-registration claim false. To confirm the release, its agents retrieved the released repositories' metadata, file listings and README files; none opened OpenAI's Lean files or the committee's assessment. Its confirmed findings were applied before deposit. Its result record is not included, because it quotes the private details it was searching for.
- **Post-release wording check.** 21 subagents (four checkers for facts, consistency, honesty and privacy, and 17 adversarial verifiers), instructed not to open OpenAI's Lean code or the assessment: about 3.8×10^6 tokens, 394 tool calls, about 14 minutes. Its confirmed findings were applied before deposit.

The orchestration scripts, which contain every subagent prompt, are included in the repository (`dev/workflows/`).

Verification and its limits.

- Every result in Table 3 is checked by Lean's kernel when the project is built, and `Audit.lean` reproduces the axiom report. No independent replay (for example with `lean4checker` or Comparator) has yet been run.
- Every bibliography entry was checked by an AI verifier for existence and bibliographic metadata against arXiv, Crossref or the cited URL. This does not mean that the cited content was read. Entries that could not be confirmed are flagged.
- Natural-language claims were reviewed only by AI agents, all instances of the same model family, whose judgements may be correlated; errors may remain.

Authorship, competing interests and funding. AI systems are not listed as authors, in line with prevailing publication norms; responsibility rests with the human author. The author declares no employment by, funding from, or financial interest in OpenAI, Anthropic or any other AI developer, and received no funding for this work. Use of the AI systems was paid for by the author through a commercial subscription. Anthropic, whose models were used, had no role in the design, conduct or reporting of this work.

10 Data and code availability

The complete package is at https://github.com/betterfuture2030/yang-mills-lean and in its Zenodo archive, which is built from the tagged Git commit:

- the Lean project (`lean/YangMills`);
- the problem description (`PROBLEM.md`);
- the eleven library notes, the completeness critique and the bibliography (`library/`);
- the plan and its five critiques (`plan/`);
- the workflow scripts and their run summaries (`dev/workflows/`);
- this paper (`paper/`).

Third-party PDFs are not redistributed. A manifest with arXiv identifiers and a download script (`scripts/fetch_pdfs.sh`) is provided instead.

Statement-file hashes. SHA-256 of the statement files of version 0.2.0:

```
lean/YangMills/YangMills/OS.lean
  36b5f3695467ed0206d12a70c0705b669d34cf9df9695982ef521eecfd546a33
lean/YangMills/YangMills/Lattice.lean
  f5807c28379cfcea5175e0fc14205b590792f89e3f7b10613cf6b77e1c1637ce
lean/YangMills/YangMills/SU.lean
  876d8e2c064c30c71a890bd243b1e5df4e91c994b7a1d4824eb3bfebe7dc8c1e
lean/YangMills/YangMills/Statement.lean
  9cbfec13626e1c3eab5f68d91a598f50765f6c3b9a3d45b5ecece2dc29a5bd05
lean/YangMills/YangMills/OpenProblems.lean
  99751a4eb989696b4c17c1eb4148643480f5786451e338df3b71f893746c009b
```

To reproduce the audit:

```
cd lean/YangMills && lake exe cache get && lake build
lake env lean YangMills/Audit.lean
```

Text is licensed under CC BY 4.0 and Lean code under Apache-2.0.

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